Contrastive Chain-of-Thought Prompting

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Abstract

Despite the success of chain of thought in enhancing language model reasoning, the underlying process remains less well understood. Although logically sound reasoning appears inherently crucial for chain of thought, prior studies surprisingly reveal minimal impact when using invalid demonstrations instead. Furthermore, the conventional chain of thought does not inform language models on what mistakes to avoid, which potentially leads to more errors. Hence, inspired by how humans can learn from both positive and negative examples, we propose contrastive chain of thought to enhance language model reasoning. Compared to the conventional chain of thought, our approach provides both valid and invalid reasoning demonstrations, to guide the model to reason step-by-step while reducing reasoning mistakes. To improve generalization, we introduce an automatic method to construct contrastive demonstrations. Our experiments on reasoning benchmarks demonstrate that contrastive chain of thought can serve as a general enhancement of chain-of-thought prompting.[1]

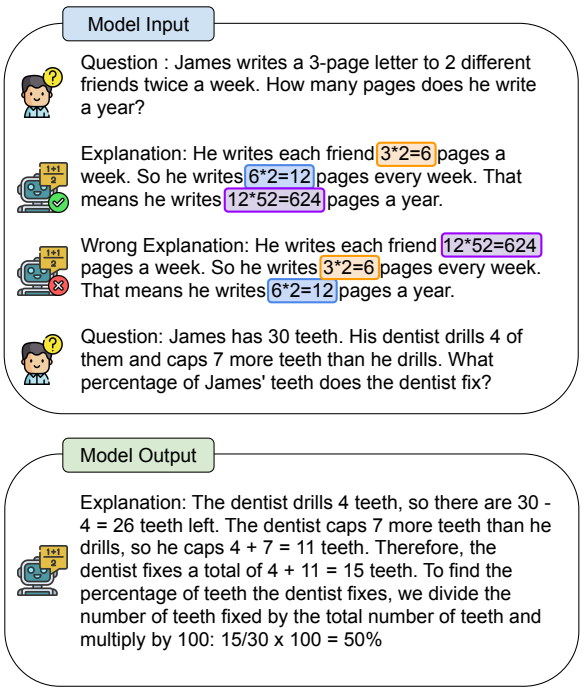

Figure 1: Example of contrastive chain-of-thought which leverages both positive and negative demonstrations to enhance language model reasoning.

1 Introduction

With the trend of large language models (LLMs), massively scaling the model size has enabled greater generalization (Brown et al., 2020) and the emergent ability to perform new tasks when given suitable prompts (Wei et al., 2022a). However, solely increasing the model size cannot solve complex reasoning tasks (Rae et al., 2022). To this end, chain-of-thought prompting was proposed to unlock the reasoning ability of LLMs by generating intermediate reasoning steps (Wei et al., 2022b). In practice, most methods based on chain of thought leverage in-context learning (Brown et al., 2020)by prompting the model with demonstrations of the input, chain-of-thought, and output (Chu et al., 2023).

However, despite its success, we lack a thorough understanding of the chain of thought (Cooper et al., 2021). For example, it was shown that even demonstrations with invalid reasoning can lead to similar performance compared to valid demonstrations (Wang et al., 2023)[2]. Hence, it is not clear how language models learn to reason effectively based on the chain-of-thought demonstrations. On the other hand, mistakes in the intermediate steps can compound and derail the reasoning process

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[1]Our code implementation will be released at https://github.com/DAMO-NLP-SG/contrastive-cot

[2]Note that while chain-of-thought can be performed in a zero-shot fashion with prompts, we focus on the few-shot setting, as it was originally proposed in Wei et al. (2022b).

(Ling et al., 2023). Any potential error in the reasoning process not only affects the accuracy of the final result but also undermines the trustworthiness of the language model (Turpin et al., 2023). Thus, it is also important to reduce mistakes in intermediate reasoning steps.

To address the challenges of chain of thought, we are inspired by how humans can learn from positive as well as negative examples. For instance, when solving a complex task where the intermediate steps are not well-defined, it is useful to learn the correct steps from positive demonstrations, as well as avoiding faults in negative demonstrations. Hence, we propose contrastive chain of thought, which provides both positive and negative demonstrations to enhance the reasoning of language models. Naturally, this raises the question of how to design effective negative demonstrations, as well as whether they can be generalized to diverse tasks. Through our analysis of multiple invalid reasoning types, we design a simple and effective method that can automatically generate contrastive demonstrations from existing valid reasoning chains. Furthermore, as contrastive chain-of-thought is task-agnostic and compatible with methods such as self-consistency (Wang et al., 2022), we believe that it can serve as a general enhancement of chain of thought.

To measure the effectiveness of contrastive chain of thought, we present evaluations on a wide range of reasoning benchmarks, and find significant benefits. Notably, compared to conventional chain of thought, we observe improvements of 9.8 and 16.0 points for GSM-8K (Cobbe et al., 2021) and Bamboogle (Press et al., 2023) respectively when using GPT-3.5-Turbo[3], a widely used LLM. Further analysis of the reasoning chains generated from our method also shows significant reduction in errors.

In summary, our main contributions include: (1) We analyse various invalid reasoning types and find that combining positive and negative demonstrations generally boost the effectiveness of chain-of-thought. (2) Based on the analysis above, we propose contrastive chain of thought to enhance language model reasoning. To improve generalization, we also propose an automatic method to construct contrastive demonstrations. (3) Evaluations on multiple reasoning benchmarks demonstrate significant improvements compared to conventional chain of thought.

[3] https://platform.openai.com/docs/models

2 Preliminary Study: Effect of Different Types of Contrastive Demonstrations

While chain of thought (CoT) prompting has enhanced the reasoning of large language models, it remains less well understood. For instance, while sound reasoning seems intuitively important to effective chain of thought, previous work has shown that there is little effect when using invalid demonstrations. On the other hand, previous works in contrastive learning (Khosla et al., 2020) and alignment (Ouyang et al., 2022) have demonstrated how language models can learn more effectively from both valid and invalid examples. Hence, we conduct a preliminary study with the following research question: **Can invalid reasoning demonstrations be instead used to enhance chain of thought?** Specifically, we aim to study the effect of providing chain-of-thought demonstrations in a "contrastive" manner, i.e., demonstrations containing both valid and invalid rationales.

2.1 Components of Chain of Thought

Compared to standard prompting with in-context demonstrations (Brown et al., 2020), chain-of-thought (CoT) prompting (Wei et al., 2022b) includes a rationale for each demonstration example. Each rationale consists of a series of intermediate reasoning steps, guiding the language model to solve tasks in a step-by-step manner. Following the formulation of (Wang et al., 2023), we identify two distinct components of each CoT rationale:

- Bridging objects are the symbolic items that the model traverses in order to reach the final solution. For example, the objects could be numbers and equations in arithmetic tasks, or the names of entities in factual tasks.

- Language templates are the textual hints that guide the language model to derive and contextualize the correct bridging objects during the reasoning process.

2.2 What is Invalid Chain of Thought?

Given the distinct components of chain of thought, we are now able to systematically identify the aspects which lead to invalid rationales. Concretely there are two main aspects which are applicable to both the language and object components:

- **Coherence** refers to the correct ordering of steps in a rationale, and is necessary for successful chain of thought. Specifically, as chain

Prompting Method	Arithmetic Reasoning Example Question: Leah had 32 chocolates and her sister had 42. If they ate 35, how many pieces do they have left in total?	Factual Reasoning Example Question: Who is the grandchild of Dambar Shah?
Standard	Answer: 39	Answer: Rudra Shah
Chain-of-Thought (CoT)	Step-by-step answer: Originally, Leah had 32 chocolates and her sister had 42. So in total they had 32 + 42 = 74. After eating 35, they had 74 - 35 = 39 pieces left in total. The answer is 39.	Step-by-step answer: Dambar Shah (? - 1645) was the father of Krishna Shah. Rudra Shah was the child of Krishna Shah (? - 1661). So the final answer (the name of the grandchild) is: Rudra Shah.
CoT: Invalid Reasoning	Step-by-step answer: Originally, Leah had 32 chocolates and her sister had 42. So her sister had 42 - 32 = 10 chocolates more than Leah has. After eating 35, since 10 + 35 = 45, they had 45 - 6 = 39 pieces left in total. The answer is 39.	Step-by-step answer: Dambar Shah (? - 1645) was the king of the Gorkha Kingdom. The Gorkha Kingdom was established by Prince Dravya Shah. Dravya Shah has a child named Rudra Shah. So the final answer (the name of the grandchild) is: Rudra Shah.
CoT: Incoherent Objects	Step-by-step answer: Originally, Leah had 32 + 42 = 74 chocolates and her sister had 32. So in total they had 74 - 35 = 39. After eating 35, they had 42 pieces left in total. The answer is 39.	Step-by-step answer: Krishna Shah was the father of Rudra Shah. Dambar Shah (? - 1645) was the child of Krishna Shah (? - 1661). So the final answer (the name of the grandchild) is: Rudra Shah.
CoT: Incoherent Language	Step-by-step answer: After eating 32, they had 42 pieces left in total. Originally, Leah had 32 + 42 = 74 chocolates and her sister had 35. So in total they had 74 - 35 = 39. The answer is 39.	Step-by-step answer: Dambar Shah (? - 1645) was the child of Krishna Shah. Rudra Shah was the father of Krishna Shah (? - 1661). So the final answer (the name of the grandchild) is: Rudra Shah.
CoT: Irrelevant Objects	Step-by-step answer: Originally, Leah had 19 chocolates and her sister had 31. So in total they had 19 + 31 = 50. After eating 29, they had 50 - 29 = 21 pieces left in total. The answer is 21.	Step-by-step answer: Metis Amando was the father of David Amando. Randall Amando was the child of David Amando. So the final answer (the name of the grandchild) is: Randall Amando.
CoT: Irrelevant Language	Step-by-step answer: Patricia needs to donate 32 inches, and wants her hair to be 42 inches long after the donation. Her hair is 35 inches long currently. Her hair needs to be 32 + 42 = 74 inches long when she cuts it. So she needs to grow 74 - 35 = 39 more inches. The answer is 39.	Step-by-step answer: The husband of Dambar Shah (? - 1645) is Krishna Shah. Krishna Shah (? - 1661) has a brother called Rudra Shah. So the final answer (the name of the brother-in-law) is: Rudra Shah.

Language Component Object Component Invalid Component (Reasoning / Language / Object)

Figure 2: Categorization of invalid chain-of-thought examples, following Wang et al. (2023).

of thought is a sequential reasoning process, it is not possible for later steps to be preconditions of earlier steps.

- **Relevance** refers to whether the rationale contains corresponding information from the question. For instance, if the question mentions a person named Leah eating chocolates, it would be irrelevant to discuss a different person cutting their hair.

In addition, following Wang et al. (2023), we include invalid reasoning as a category of invalid chain of thought, which is neither incoherent nor irrelevant, but contains logical mistakes. Hence, we aim to study the five main categories of invalid chain-of-thought, as shown in Figure 2.

2.3 Experimental Setup

To conduct the experiments for the preliminary study, we leverage the GSM8K (Cobbe et al., 2021) and Bamboogle (Press et al., 2023) datasets for arithmetic and factual reasoning respectively. We use the OpenAI Chat Completions API[4] which is one of the most popular and well-performing language models with reasonable cost. Specifically, we use the GPT-3.5-Turbo (0301) version. To study the effect of contrastive demonstrations under various settings, we evaluate the five main invalid categories as shown in Figure 2. Note that we use 4-shot prompting for each dataset, and the chain-of-thought demonstrations are manually constructed by previous works (Wei et al., 2022b; Wang et al., 2023). To standardize the prompting process, we use a simplified chain-of-thought prompt format, as shown in Figure 1.

2.4 Preliminary Results

Based on the preliminary results in Table 1, we observe significant gains across all invalid rationale categories compared to conventional chain-of-thought. Notably, leveraging chain of thought with contrastive demonstrations containing incoherent objects yields the highest average performance

[4] https://platform.openai.com/docs/api-reference